Pittala Sree Sai Pavan

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B.Tech – Electronics and Communication Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech.	Indian Institute of Technology Guwahati	7.13 (Current)	2024 – Present
Senior Secondary	Telangana Board of Intermediate Education	96.2%	2024
Secondary	Telangana Board of Secondary Education	9.7	2022

Projects

- **MAC-16: Custom 16-Bit AI-Accelerated RISC Processor** *May 2026*
  Digital System Design & Computer Architecture Github
  - Designed a **16-bit single cycle RISC processor** in **Verilog**, custom built and optimized for edge AI workloads.
  - Developed a **dual lane SIMD MAC hardware unit** executing two parallel 8-bit math operations per clock cycle.
  - Implemented **zero overhead hardware looping** to eliminate all branch and jump penalties during execution loops.
  - Achieved a **6.4x speedup**, reducing an eighteen element dot product execution from **51 to 8 clock cycles**.
  - Verified processor logic and accumulator writebacks using cycle-accurate **behavioral simulation** in **Xilinx Vivado**.
- **Two-Stage Miller-Compensated Operational Transconductance Amplifier** *April 2026*
  Analog Integrated Circuit Design LTspice
  - Designed a **two-stage OTA** in **TSMC 0.18 µm CMOS** achieving **71 dB DC gain** and **GBW ≈ 14 MHz** at 1.8 V supply.
  - Applied **Miller compensation** with **zero-cancellation**, yielding **88° phase margin** — 28° above the stability criterion.
  - Sized six **MOSFETs** via **gm/ID methodology** ($V_{ov} = 200\,\text{mV}$); verified **closed-loop gain of 2×** through **step response**.
  - Implemented a **10× NMOS current mirror** for tail biasing, achieving **100 µA tail current** at **0.38 mW** total power.
- **Traffic Signal Controller on FPGA** *March 2026*
  Digital Logic Design Verilog HDL
  - Designed a **2-state synchronous FSM** in **Verilog HDL** for a 2-path traffic controller with **5-second** phase sequencing.
  - Implemented a **clock divider** module scaling the system clock down to **1 Hz** for accurate synchronous state transitions.
  - Synthesised and deployed onto a **Zedboard (Zynq-7000) FPGA** via **Vivado**; validated **LED sequencing** on hardware.
- **Automated Circuit Analyser** *February 2026*
  Computational Electrical Engineering Python, MATLAB
  - Built a **Python** circuit solver using **NetworkX** to model circuits as derected graphs for systematic **KCL/KVL** analysis.
  - Generated **Cut Set** and **Tie Set** matrices automatically from **spanning tree** to formulate independent circuit equations.
  - Solved **Laplace-domain** impedance matrices via **SymPy**, computing s-domain voltages and currents across all branches.
  - Applied **inverse Laplace transforms** to obtain **time-domain transient responses**, visualised using **Matplotlib** for all nodes.

Technical Skills

- **Hardware & Simulation**: LTspice, Verilog HDL, MATLAB, Simulink*
- **Programming Languages**: C, C++, Python, HTML, CSS, JavaScript
- **Tools/Libraries**: NumPy, Pandas, Matplotlib, sympy, Networkx, Git, GitHub
- **Operating Systems**: Windows, Linux ** Elementary proficiency*

Relevant Coursework

- **Core Electronics**: Basic Electronics, Digital Circuits (Theory & Lab), Circuit Theory, Signals and Systems, Analog Circuits, Control Systems
- **Mathematics**: Linear Algebra, Calculus, Differential Equations, Real Analysis, Complex Analysis, Probability and Random Processes

Achievements

- **Competitive Programming**, Max Rating **1404** on Codeforces (**Specialist**); Handle **- ssp_09** *2025 – 26*
  - Ranked **1320** globally among 34,550+ participants in Codeforces Round 1090 (Div. 4).
  - Achieved Global Rank **1994** among 21,113+ participants in Codeforces Round 1083 (Div. 2).
- **JEE Advanced 2024**, Ranked in the top **2.53%** among **1.8 lakh**+ shortlisted candidates across India *2024*
- **TS EAPCET 2024**, Ranked in the top **0.87%** among **2.4 lakh**+ candidates across Telangana State *2024*

Extracurricular Activities

- **CoS, NSS Volunteer**, NSS Cell, IIT Guwahati: Contributed 30+ hours to community service activities, including village visits, face painting, and engagement programs. *Jan. 2026 - present*
- **Junior Web Developer**, Students Web Committee — frontend dev on institute projects *Oct. 2025 - Present*
- **PR & Branding Associate**, Alcheringa — Northeast India's largest cultural fest *Nov. 2024 - Feb. 2025*